

Department Overview

- **Department Name:** Department of Geological Engineering
- **Faculty:** Faculty of Engineering & Architecture
- **Accreditation:** Officially accredited by the **Pakistan Engineering Council (PEC)**.
- **History:** Initiated in 2008 as the first degree-awarding Geological Engineering program under the engineering faculty at BUITEMS. It was established to address the critical need for hidden natural resource exploration and economic development in Balochistan, which is Pakistan's largest and most mineral/hydrocarbon-rich province (with roughly 90% of its area historically unexplored).
- **Core Technical Domains:** Analyzing the engineering behavior of rock, soil, and geotechnics against constructed projects, and guiding mineral and hydrocarbon discovery.
- **Career Sectors:** High demand in the energy and construction sectors. Graduates find roles as Geotechnical Engineers, Geologists, Construction Consultants, or Civil Contractors across public and private organizations. Key employment fields include petroleum/gas organizations, atomic energy, the Irrigation Department, academic institutions, and Natural Energy Resources Exploration.

Mission

The Department of Geological Engineering is committed to train engineers with advanced knowledge and contemporary skills, while promoting a culture of research, equipping students with the art of living to achieve success in their future professional careers not only in Pakistan but throughout the world.

Program Educational Objectives (PEOs)

Graduates are expected to demonstrate the following attributes 3 to 5 years post-graduation:

1. The ability to apply technological and engineering fundamentals for the solution of geotechnical and environmental problems that are related to Geological processes.
2. The ability to exhibit effective communication, teamwork, leadership skills, and technical competence enabling them to excel in their professional careers as well as in higher studies.
3. The ability to fulfill ethical, professional, and social responsibilities in their professional lives and community services.

Bachelor of Science in Geological Engineering (BS Geological Engineering)

Admission Requirements

- **Academic Background:** F.Sc (Pre-Engineering / Pre-Medical / ICS) from any recognized

board with at least **60% marks**, OR a Diploma of Associate Engineer (DAE) in Civil, Land and Mine Surveying, or Mining securing a minimum of **60% marks**.

- **Testing:** Candidates must successfully qualify for the university admission test.

Degree Requirements

- **Total Credit Hours:** 136
- **Total Courses:** 46
- **Minimum Academic Standing:** CGPA $\geq$ 2.0
- **PLO Requirement:** The program adopts the 12 expected standard engineering graduate attributes defined in the PEC Accreditation Manual 2014 (Engineering Knowledge, Problem Analysis, Design/Solutions, Investigation, Modern Tool Usage, Engineer & Society, Sustainability, Ethics, Teamwork, Communication, Project Management, and Lifelong Learning).

Curriculum Plan (Program Schema by Semester)

Note: Credit hour structures denote (Theory + Lab).

Semester	Course Code	Course Title	Credit Hours
1st	ENGG-201 / ENGG-111	Engineering Drawing & Graphics / Basic Electrical & Electronics	1+1 / 2+1
	GEOE-203 / CHE-102	Physical Geology / Applied Chemistry	3+1 / 2+0
	MATHP-118 / HUM-101	Calculus & Analytical Geometry / Islamic Studies or Ethics	3+0 / 2+0
	HUM-163	Functional English	2+0
2nd	GEOE-305 / GEOE-202	Mineralogy & Petrology / Intro to Geological	2+1 / 2+0

		Engineering	
	MATHP-214 / PHY-205	Differential Equations / Applied Physics	3+0 / 2+0
	Hum-268 / HUM-102	Communication Skills / Pak-Studies	2+0 / 2+0
	IT-102 / ENIRON-305	Intro to ICT / Occupational Health & Safety	2+* / 1+0
3rd	ENGG-215 / MINE-201	Applied Thermodynamics / Stratigraphy & Structural Geology	2+1 / 2+1
	MATHP-111 / ENGG-206	Linear Algebra / Fluid Mechanics	3+0 / 3+1
	CS-112 / ENGG-241	Computer Programming / Engineering Economics	2+1 / 2+0
4th	GEOE-206 / GEOE-208	Mechanics of Material / Surveying	3+1 / 2+2
	CS-115 / GEOE-312	Computer Science & Numerical Analysis / Hydrogeology	2+0 / 3+1

	STAT-101 / HUM-276	Probability & Statistics / Tech Report Writing & Pres. Skills	2+0 / 2+0
5th	GEOE-302 / GEOE-401	Engineering Geology / Intro to Geophysical Exploration Tech	2+1 / 3+1
	GEOE-311 / MINE-307	Geotechnical Engineering-I / Explosives Engineering	3+1 / 2+1
	MGMT-211 / SOC-301	Engineering Management / Sociology for Engineers	2+0 / 2+0
6th	GEOE-313 / MINE-401	Petroleum Geology / Rock Mechanics	2+1 / 3+1
	GEOE-403 / GEOE-316	Earth Quake Seismology & Risk Assessment / GIS & Remote Sensing	2+1 / 2+1
	GEOE-417	Environmental Geological Engineering	2+1
7th	— / MINE-404	Major Elective I / Tunneling & Excavation	3+1 / 3+1

		Engineering	
	— / —	Major Elective II / Drilling Engineering	2+1 / 2+1
	GEOE-410L	Project - I	0+2
8th	— / —	Major Elective III / Major Elective IV	3+1 / 2+1
	MGMT-201 / GEOE-410L	Entrepreneurship / Project - II	3+0 / 0+4
		Grand Total	136 Credit Hours

**Note: The raw text leaves the lab component indicator incomplete for IT-102 in the layout list.*

Major Electives Curriculum (5th Board of Studies)

The program offers 24 elective courses partitioned into three distinct professional specializations under the current curriculum framework. Students choose 4 courses in total from one preferred stream (2 courses in the 7th semester, and 2 courses in the 8th semester):

- **GRE:** Geotechnical and Rock Engineering
- **NERE:** Natural Energy Resource Exploration
- **GEE:** Geo-Environmental Engineering

Specialization Streams Course Options

- **Major Elective-I (Semester 7 — 3+1 Credits):**
 - *GRE:* Geotechnical Engineering-II **OR** Statistical Methods in Geology and Engineering
 - *NERE:* Seismic Data Processing & Interpretation **OR** Ground water Resource Engineering
 - *GEE:* Environment Geology & Waste Management **OR** Environmental Aspects of Mining
- **Major Elective-II (Semester 7 — 2+1 Credits):**
 - *GRE:* Geotechnical Site Investigation **OR** Fluvial Processes
 - *NERE:* Reservoir Engineering **OR** Surficial Processes, Sedimentation and Stratigraphy
 - *GEE:* Risk Assessment in Environmental Studies **OR** Exploration and Environmental Geochemistry
- **Major Elective-III (Semester 8 — 3+1 Credits):**

- *GRE*: Pavement and Foundation Engineering **OR** Geomorphology and Terrain Analysis
- *NERE*: Petrophysics and Well Logging **OR** Petrogenesis and Metallogenesis
- *GEE*: Environmental Soil Science **OR** Site Remediation Engineering
- **Major Elective-IV (Semester 8 — 2+1 Credits):**
 - *GRE*: Ground Improvement and Geosynthetics **OR** Analysis of Rock Structures
 - *NERE*: Production Engineering **OR** Carbonate Sedimentology
 - *GEE*: Environmental Data Analysis **OR** Recycling and Sustainable Engineering